

Spivak Chapter 4 Problems

January 2, 2026

1 Problem 1

Indicate on a straight line the set of all x satisfying the following conditions. Also name each set, using the notation for intervals.

1. $|x - 3| < 1$.
 $2 < x < 4$,
 $(2, 4)$.
2. $|x - 3| \leq 1$.
 $2 \leq x \leq 4$,
 $[2, 4]$.
3. $|x - a| < \epsilon$.
 $(a - \epsilon, a + \epsilon)$.
4. $|x^2 - 1| < \frac{1}{2}$.
 $(-\sqrt{\frac{3}{2}}, -\sqrt{\frac{1}{2}}) \cup (\sqrt{\frac{1}{2}}, \sqrt{\frac{3}{2}})$.
5. $\frac{1}{1+x^2} \geq \frac{1}{5}$.
 $[-2, 2]$.
6. $x^2 + 1 \geq 2$.
 $(-\infty, -1] \cup [1, \infty)$.
7. $(x + 1)(x - 1)(x - 2) > 0$.
 $(-1, 1) \cup (2, \infty)$.

2 Problem 2

There is a very useful way of describing the points of the closed interval $[a, b]$.

1. First consider the interval $[0, b]$, for $b > 0$. Prove that if x is in $[0, b]$, then $x = tb$ for some t with $0 \leq t \leq 1$. What is the significance of the number t ? What is the midpoint of the interval $[0, b]$?

Let $t = \frac{x}{b}$. Because $x \in [0, b]$, $0 \leq x \leq b$. Thus, $0 \leq \frac{x}{b} \leq 1$, so $0 \leq t \leq 1$. Then, $tb = \frac{x}{b}b = x$.

2. Now prove that if x is in $[a, b]$, then $x = (1 - t)a + tb$ for some t with $0 \leq t \leq 1$.

Because $x \in [a, b]$, $a \leq x \leq b$. Let $t = \frac{x-a}{b-a}$. Then, $0 \leq t \leq 1$, and $x = a + t(b - a)$, as required.

3. Prove, conversely, that if $0 \leq t \leq 1$, then $(1 - t)a + tb$ is in $[a, b]$.

If $(1 - t)a + tb \in [a, b]$, then $a \leq a - at + tb \leq b$. Then, $0 \leq t(b - a) \leq b - a$. This condition is satisfied for $0 \leq t \leq 1$.